

SCOPE OF WORK

Smart Student Productivity Platform (SSPP)

Centralized Academic Productivity & Management Platform

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Project Team

Name	Role
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1. Project Overview

1.1 Executive Summary

The Smart Student Productivity Platform (SSPP) is a centralized, web-based academic productivity and management platform designed to bring essential academic activities into a single integrated environment for Students, Faculty, Heads of Department (HOD) and authorized Administrators.

The platform aims to simplify academic management by integrating academic activities, communication, productivity tools, online classes, AI-assisted doubt resolution and role-based reporting into one system.

SSPP will be delivered as a responsive web application using HTML5, CSS3 and JavaScript, with Firebase serving as the primary backend platform.

1.2 Business Problem

Problem Area	Current Challenge	Expected Improvement
Academic Information	Information distributed across multiple systems	Centralized academic information
Assignments	Separate tools and communication channels	Centralized assignment management
Communication	Multiple communication channels	Integrated communication
Attendance & Results	Separate academic systems	Centralized academic records
Student Productivity	Tasks and schedules managed separately	Integrated tasks, calendar and productivity tools
Doubt Resolution	No centralized first-level assistance	AI-assisted doubt resolution
Department Monitoring	Limited centralized visibility	HOD dashboards and reports
Administration	Repetitive administrative activities	Centralized administration and role-based access

1.3 Proposed Solution

Component	Proposed Solution
Academic Management	Centralized management of courses, assignments, assessments, attendance and examinations
Student Productivity	Tasks, calendar, study groups and exam-focused study
Communication	Student–faculty and departmental communication
AI Support	AI-assisted first-level academic doubt resolution
Online Learning	Google Meet integration
Administration	User, academic structure, role and permission management
Security	Authentication and role-based authorization
Platform	Responsive web application

1.4 Business Objectives

The project aims to:

1. Centralize academic activities for students and faculty.
2. Improve student productivity around schedules, deadlines and personal tasks.
3. Improve communication between students, faculty and administrators.
4. Reduce repetitive administrative work for faculty.
5. Provide AI-assisted first-level academic doubt resolution.
6. Improve HOD visibility into student performance, faculty workload and allocation.

7. Support configurable academic programs such as B.Tech, BBA and MBA.
8. Improve accessibility of academic data for authorized users.
9. Improve security through authentication, authorization and role-based access.
10. Provide scalable infrastructure with minimal administrative overhead.

2. Scope of Work

Module	Feature Coverage	Count
Student Portal	Dashboard, academics, courses, resources, assignments, assessments, projects, examinations, attendance, results/feedback, productivity, tasks, calendar, communication, online classes, notifications, AI Doubt Solver, group study rooms, exam-focused study, profile & security	20
Faculty Portal	Dashboard, academic mapping, content management, assignment creation/evaluation, attendance, assessment/examination evaluation, online meeting scheduling, communication	9
HOD Portal	Dashboard, faculty allocation, student monitoring, faculty monitoring, timetable management, academic approval, departmental reports	7
Administration Layer	User management, academic structure, role/permission management, system configuration, Firebase/integration configuration, audit/security oversight	Supporting Layer
Platform	Responsive web application for desktop and mobile browsers	—

2.2 Out of Scope

The following items are outside the current project scope:

- Native Android application

- Native iOS application
- Finance and payment management
- Hostel management
- Transport management
- Other non-academic administration modules
- Learning-content authoring tools beyond uploading and organizing existing materials
- Third-party LMS integrations other than Google Meet

2.3 Target Users

User	Primary Responsibility / Usage
Students	Academic activities, productivity, communication and study support
Faculty	Academic delivery, evaluation, attendance, content and communication
HOD / Department Heads	Departmental monitoring, allocation, approvals and reporting
Administrators / Super Administrators	User management, configuration, governance and security

3. Business & User Requirements

3.1 Stakeholders

Stakeholder	Responsibility / Interest
Business / Institution Sponsor	Overall project sponsorship and approval
Students	Primary users of academic and productivity features
Faculty	Academic delivery and student evaluation

HOD / Department Heads	Departmental operations and academic monitoring
Administrators	Platform-wide configuration
Development Team	Design and development of the platform

The current SOW identifies these stakeholder groups and their responsibilities.

3.2 User Roles

Role	Responsibility
Super Admin	Platform-wide administration, users, roles, permissions, settings and audit
Administrator	Academic structure and user administration
HOD	Academic operations within assigned department
Faculty	Academic activities for assigned subjects and sections
Student	Personal academic activities and productivity

3.3 Role-to-Portal Access

Capability	Super Admin	HOD	Faculty	Student
Student Portal	Support / Oversight	Department Oversight	—	Full Access
Faculty Portal	Support / Oversight	Department Oversight	Full Access	—
HOD Portal	Full / Support	Full Access	—	—
User Management	Full	Department Scope	—	—
Academic Structure	Full	Department Scope	View Assigned	View
Role Management	Full	Limited	—	—

Reports	Platform	Department	Assigned Classes	Personal
Audit Logs	Full	Department	Limited	Personal Activity
System Configuration	Full	—	—	—

4. Functional Requirements

Module	Functional Scope
Student Portal	Academic dashboard, courses and resources, assignments and assessments, examinations and results, attendance, productivity and scheduling, communication, online classes, notifications, AI-assisted doubt solving, collaborative study and profile management.
Faculty Portal	Academic dashboard, subject and content management, assignment creation and evaluation, attendance, assessments and examinations, online meeting management and communication.
HOD Portal	Department dashboard, faculty allocation, student and faculty monitoring, timetable management, academic approvals and departmental reporting.
Administration layer	User management, academic structure management, role and permission management, system configuration, integrations and audit/security oversight.

4.1 Functional Coverage

Area	Primary Purpose
Academic Management	Manage courses, subjects, resources, assignments, assessments, attendance and examinations.

Productivity	Support student tasks, schedules, study activities and exam preparation.
Communication	Facilitate communication between students, faculty and academic administrators.
Collaboration	Support online classes and group study activities.
AI Support	Provide first-level academic doubt resolution through the AI Doubt Solver.
Monitoring & Reporting	Provide faculty and HOD-level monitoring, approvals and reports.
Administration & Security	Manage users, roles, permissions, academic structure, configuration and audit activities.

5. Non-Functional Requirements & Business Rules

5.1 Non-Functional Requirements

Category	Requirement
Security	Secure authentication, role-based authorization, Firebase Security Rules, HTTPS and protected backend credentials
Performance	Optimized database queries, file loading, dashboard retrieval and Firebase operations
Scalability	Infrastructure should support growth in users, departments, records and AI requests
Availability	Platform should be accessible during normal institutional operating periods, subject to service availability
Usability	Clear navigation, role-specific dashboards, responsive layouts and meaningful system states

Data Integrity	Academic and user data should remain consistent and controlled
Auditability	Critical actions should be recorded for audit purposes
Reliability	Graceful handling of network, upload, notification and authorization failures

5.2 Key Business Rules

1. Public unrestricted self-registration is not permitted.
2. User accounts are provisioned by authorized administrators.
3. Access is restricted according to the user's authorized academic scope.
4. Group study rooms are limited to a maximum of five participants.
5. A sixth participant cannot join a full study room.
6. Marks and results remain hidden until officially published.
7. Academic profile fields such as Program, Semester and Section are controlled by authorized administrators.
8. AI Doubt Solver requests must be routed through an authenticated backend mechanism.
9. Deactivated users cannot authenticate.
10. Required records are retained for audit purposes.

6. Proposed Solution & Technology

6.1 Solution Approach

SSPP will be delivered as a **responsive web application** supporting the following academic hierarchy:

Institution → Department → Program → Semester → Section → Subject → Faculty / Student

The platform will provide separate role-based experiences for Students, Faculty, HODs and Administrators while using a shared Firebase backend.

The solution is intended to centralize academic operations while maintaining appropriate access controls for each user role.

6.2 High-Level Architecture

Layer	Component	Purpose
Presentation	HTML5 / CSS3 / JavaScript	Responsive user interface
Authentication	Firebase Authentication	User authentication
Database	Cloud Firestore	Primary application data
Real-Time	Firebase Realtime Database	Real-time functionality where required
Backend	Firebase Cloud Functions	Backend logic and AI request routing
Storage	Cloudinary	File storage
Notifications	Firebase Cloud Messaging	Notifications
Hosting	Firebase Hosting	Application deployment
Security	Firebase Security Rules	Data and access security
Online Classes	Google Meet	Online meetings

6.3 Technology Stack

Layer	Technology / Service
Frontend	HTML5, CSS3, JavaScript
Authentication	Firebase Authentication
Primary Database	Cloud Firestore
Real-Time Features	Firebase Realtime Database

File Storage	Cloudinary
Backend Logic / AI Routing	Firebase Cloud Functions
Hosting	Firebase Hosting
Notifications	Firebase Cloud Messaging
Security	Firebase Security Rules
Online Meetings	Google Meet Integration

7. Deliverables & Acceptance Criteria

7.1 Project Deliverables

The project deliverables will include:

1. Approved Business Requirements Document (BRD).
2. Approved Statement of Work (SOW).
3. Finalized business and functional requirements.
4. Defined user roles and access matrix.
5. Responsive Student Portal.
6. Responsive Faculty Portal.
7. Responsive HOD Portal.
8. Supporting Administration capabilities.
9. Implementation of the agreed 36 feature items.
10. Firebase integration and configuration.
11. Google Meet integration.
12. Testing and validation of critical workflows.
13. Production deployment.
14. Project documentation including BRD, SOW, PRD and technical documentation.

7.2 Acceptance Criteria

Area	Acceptance Criteria
Feature Completion	All agreed features are implemented and tested

User Workflows	Student, Faculty and HOD workflows function end-to-end
Authentication	Firebase authentication operates correctly
Authorization	Role-based access restrictions are enforced
Study Rooms	Maximum five-member capacity is enforced
Results	Results remain hidden until officially published
Audit	Critical actions are captured in audit records
Integration	Required integrations function correctly
Stakeholder Acceptance	Final requirements are formally accepted and signed off

8. Project Timeline

The project is organized into **five sprints** covering the project period from **25 August 2026 through 30 September 2026**.

Sprint	Dates	Major Activities
Sprint 1	25 Aug – 29 Aug	Project kickoff, domain research, business problem identification, proposed solution, business objectives, scope definition, stakeholder analysis, user requirements and BRD finalization.
Sprint 2	31 Aug – 5 Sep	SOW preparation, project deliverables and responsibilities, SOW timeline and milestones, SOW review/finalization, development discussion, user registration, login and authentication.
Sprint 3	7 Sep – 12 Sep	Development activities based on approved requirements.
Sprint 4	14 Sep – 19 Sep	Feature development, integration and testing.

Sprint 5	21 Sep – 30 Sep	Final integration, testing, refinement, deployment and documentation.
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9. Risks, Dependencies, Assumptions & Change Control / Sign-off

9.1 Key Risks

Risk	Impact	Mitigation
Firebase usage / cost growth	High	Monitor usage, optimize database queries and establish billing limits.
Academic-structure changes during the project	Medium	Use controlled administrator workflows for academic-structure changes.
Inaccurate AI responses	High	Apply AI guardrails and provide faculty escalation for unresolved or complex doubts.
Google Meet outages / licensing issues	Medium	Provide alternative communication and class rescheduling mechanisms.
Low user adoption	Medium	Provide onboarding, training and user guidance.
Academic-data privacy exposure	High	Enforce Firebase Authentication, role-based access and strict Security Rules.

9.2 Dependencies

The successful delivery of SSPP depends on:

- Firebase project and required service configuration.
- Firebase billing configuration where applicable.
- Google Workspace / Google Meet availability.
- Finalized academic structure.
- Faculty and student master data provided by the institution.
- Firebase Cloud Messaging configuration.
- Any required content migration from existing systems.

9.3 Assumptions

No.	Assumption
1	A Super Admin will be provisioned during initial platform setup.
2	One Administrator/HOD will manage each domain unless the structure is extended.
3	Required Firebase services will be available before development begins.
4	Google Meet will be available for online classes.
5	The academic structure will be finalized before system configuration begins.

9.4 Change Control

Step	Activity	Description
1	Document	Record the requested change and its purpose
2	Assess	Evaluate impact on scope, timeline, cost and feature baseline
3	Decision	Approve, defer or reject the requested change
4	Update	Update relevant BRD/PRD/project documentation
5	Communicate	Communicate approved changes to stakeholders and project team

9.5 Sign-off

Name	Title / Role	Signature	Date
Sayantan Ghosh	Business Sponsor (Client)		
Sakshi Shingole	Project Manager + BA		
Sweta Kadam	Scrum Master		
Riddhi Zunjarrao	Technical Lead		
Ragini Singh	Solution Architect		